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Flapping-enhanced active olfactory sampling in mosquitoes through near-field odor transport

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Abstract

Mosquitoes rely on olfactory cues for host seeking, mating, and foraging. Although extensive studies have investigated molecular and behavioral responses, the influence of flapping wing in modulating near-field odor transport and antennal sensing remains poorly understood. In this study, a three-dimensional numerical framework coupling flapping aerodynamics with passive scalar transport was developed to investigate mosquito olfaction. The results show that periodically generated vortices continuously entrain surrounding odor toward the antennae, producing stronger temporal fluctuations and higher time-averaged odor concentration. Lagrangian analysis further reveal coherent transport pathways responsible for the enhanced odor delivery. From an inversed sensing perspective, flapping introduces high nonlinearities, thus significantly reducing the accuracy of conventional Gaussian plume models for odor source localization. And we propose a new physics-guided neural network surrogate model to improve it. These findings demonstrate that flapping motion not only generate aerodynamic forces, but also enhances active olfactory sampling through near-field flow modulation, providing theoretical insights for the development of bio-inspired or odor-guided aerial robots.

1. Introduction

Active olfactory navigation is widespread across biological systems and enables animals to locate food, mates and hosts. Unlike visual cues, odor signals are transported by wind or water, producing intermittent and spatially complex chemical plumes. Animals must therefore actively extract reliable information from these highly dynamic environments. To achieve this, many species have evolved specialized morphological structures or sensing behaviours to regulate local fluid transport and enhance signal acquisition. For example, lobsters actively flick their antennules to increase odor penetration into sensory hair arrays [1], while wing flapping in flying insects periodically refreshes the air surrounding the antennae, preventing receptor saturation [2,3].

Among odor-guided insects, mosquitoes serve as an ideal model system because their host-seeking behaviour strongly depends on olfactory cues, particularly carbon dioxide. Exposure to CO_2 plumes has been shown to switch mosquitoes from passive flight to active environmental exploration [4]. Furthermore, female mosquitoes can detect CO_2 concentration as low as 103 ppm and rely on this remarkable sensitivity to locate human hosts [5]. As vectors of malaria, dengue fever and numerous other infectious diseases, mosquitoes are of considerable medical importance. Understanding how mosquitoes acquire olfactory information is therefore valuable not only for disease control but also for the design of bio-inspired sensing systems.

Previous studies on mosquito olfaction have primarily focused on three aspects: molecular sensing mechanisms, behavioural responses and systems-level modelling. Genetic studies have demonstrated that odorant receptor pathways are essential for host discrimination and responses to volatile

repellents [6]. Behavioural experiments have revealed the role of carbon dioxide in host localization [4], while recent computational models have successfully reproduced mosquito flight trajectories using Bayesian dynamical systems [7]. Despite these advances, odor cues are typically represented by prescribed concentration fields, idealized Gaussian plumes or empirical concentration gradients. Consequently, the near-field aerodynamic processes that reshape odor signals before they reach the antennae remain largely unexplored. In particular, how flapping-induced flow structures modify odor transport and influence olfactory sampling has yet to be quantitatively understood.

In this work, we develop a three-dimensional computation framework coupling flapping aerodynamics with passive scalar transport to investigate flapping-enhanced active olfactory sampling in mosquitoes. To accurately solve scalar transport near one-layer flapping wings, a scalar jump-condition correction method is first developed. The validated framework is then used to quantify how flapping-induced vortex reshape near-field scalar transport and enhance odor sampling. Finally, odor source localization is formulated as an inverse sensing problem, demonstrating that the strong nonlinearities introduced by flapping significantly changes conventional Gaussian plume models, thereby motivating a neural network inference framework. This study provides a fluid mechanical perspective on active olfactory sampling and offers design principles for bio-inspired chemical sensing and odor-guided robotic navigation.

2. Methodology

To investigate how flapping flight influences olfactory sensing in mosquitoes, a three-dimensional numerical framework coupling flapping aerodynamics with passive scalar transport was developed. The mosquito was placed in a virtual wind tunnel under a uniform inflow to simulate upwind flight. The flow field was solved using an in-house immersed boundary (IB) solver, while odor transport was modelled as a passive scalar governed by the advection–diffusion equation.

The mosquito geometry and wing kinematics were adapted from previous experimental and numerical studies [8–11]. Representative kinematic parameters for female mosquitoes were adopted, including flapping frequency of 421 Hz, stroke amplitude of $39^\circ \pm 4^\circ$, wing length of 2.5–2.8 mm (R) and forward flight speed of $U_f \approx 0.25$ m/s. Wing motion was prescribed using three Euler angles: the stroke, deviation and pitch angle, as illustrated in Fig. 1(a, b). To characterize odor sensing, thirteen virtual sampling probes were distributed along each antenna according to the experimentally identified spatial distribution of olfactory receptor neurons (Fig. 1d) [12].

The computational domain consisted of a Cartesian cube of $4R \times 4R \times 4R$, discretized using a uniform 256^3 grid. The mosquito body and wings were represented by a high-resolution triangular surface mesh. As shown in Fig. 1(c), Dirichlet boundary conditions were applied at the inlet and the upper and lower boundaries of the flow field, while no-slip boundary condition was enforced on the mosquito surface. For scalar transport, a constant-concentration odor source was positioned upstream of the mosquito using a Dirichlet boundary condition, whereas a zero normal-gradient (Neumann) boundary condition was imposed on the insect surface to represent an impermeable boundary.

The unsteady three-dimensional incompressible Navier–Stokes equations are described by:

$$\frac{\partial u}{\partial t} + u \cdot \nabla u = -\nabla p + \frac{1}{Re} \nabla^2 u + f$$

$$\nabla \cdot u = 0$$

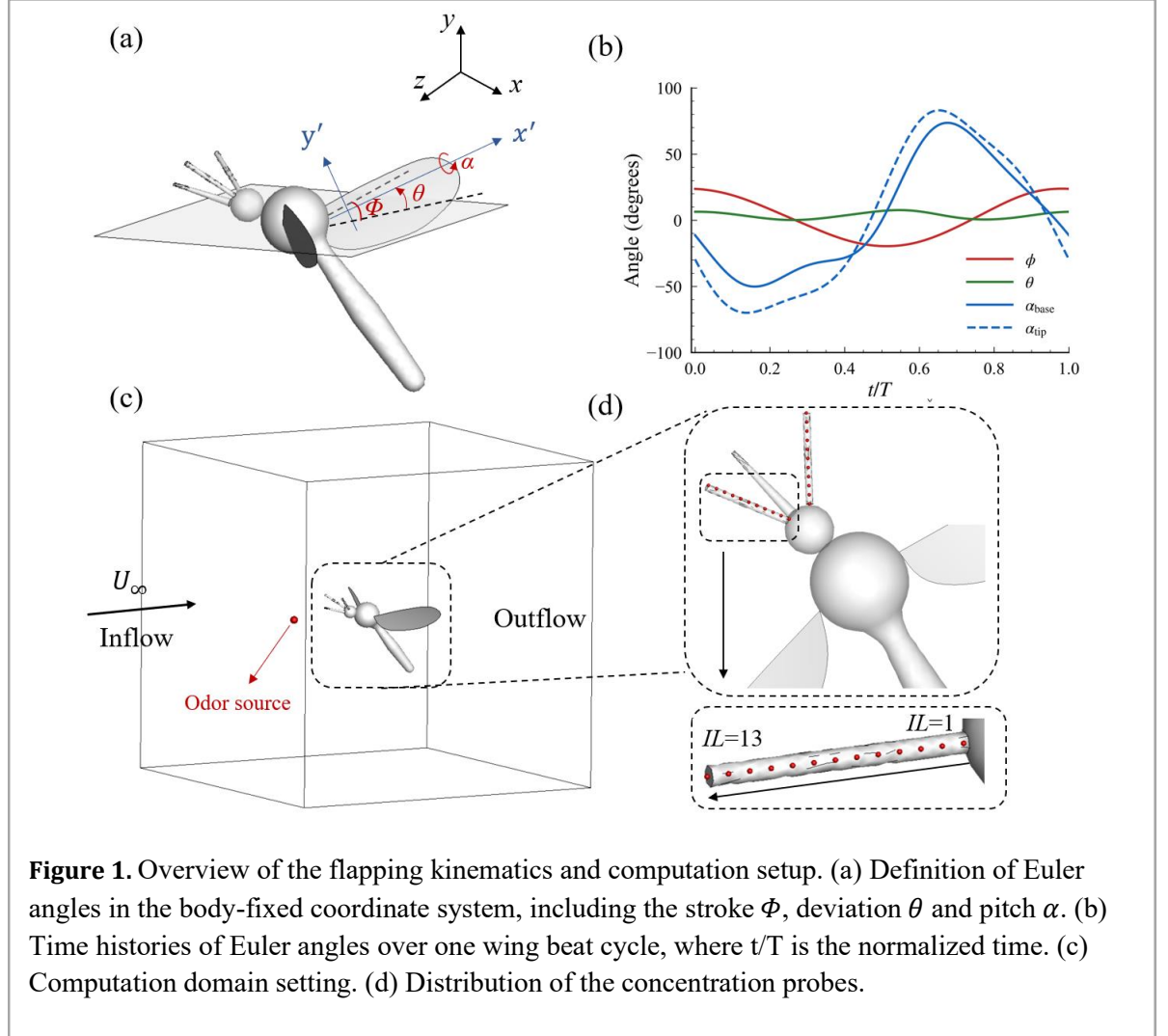
where u is the velocity; p is the pressure; Re is the Reynolds number; and f denotes the immersed boundary forcing term used to enforce the no-slip boundary condition on the moving body surface. The whole flow equations are solved using a fractional-step projection method. And the no-slip boundary condition on the moving mosquito surface is enforced using an immersed boundary forcing approach.

Odor transport is modelled as a passive scalar governed by the advection–diffusion equation,

$$\frac{\partial C}{\partial t} + u \cdot \nabla C = \frac{1}{Sc \cdot Re} \nabla^2 C$$

where C is the scalar concentration; u is the flow velocity; Re and Sc are the Reynolds and Schmidt numbers respectively.

The scalar field considers both Dirichlet and Neumann boundary conditions. A constant



concentration is prescribed at the odor source using a Dirichlet boundary condition, whereas a Neumann boundary condition is imposed on the mosquito surface to represent an impermeable boundary. However, for one-layer thin flapping wings, zero normal-gradient must be satisfied simultaneously on both sides of the immersed interface. Thus, traditional immersed boundary treatments can't accurately impose this constraint and often introduce numerical oscillations near the interface.

To overcome this limitation, a scalar jump-condition correction method is proposed for thin boundaries. Following the jump-condition formulation developed for immersed boundary flow solvers [13], the governing equations are reformulated in its weak form. Thus, the discontinuities in the scalar field and its normal derivative across the interface are represented by equivalent forcing terms. The detailed derivation is presented in Appendix A. The governing equation is therefore rewritten as

$$\frac{\partial C}{\partial t} + u \cdot \nabla C = D \nabla^2 C + f$$

where

$$f = \iint F_1 \delta(x - X) dr ds + \iint F_2 \nabla \delta(x - X) \cdot n dr ds$$

The forcing coefficients are given by

$$F_1 = [\nabla C_n] \left| \frac{\partial X}{\partial r} \times \frac{\partial X}{\partial s} \right|$$

$$F_2 = D[C] \left| \frac{\partial X}{\partial r} \times \frac{\partial X}{\partial s} \right|$$

where $[C]$ denotes the concentration jump across the interface; $[\nabla C_n]$ denotes the normal concentration gradient jump. Physically, F_1 corresponds to a monopole forcing that enforces the prescribed normal scalar flux; and F_2 corresponds to a dipole forcing that maintains the concentration discontinuity across the interface. Together, these two forcing terms provide a unified treatment for Dirichlet, Neumann, and general jump boundary conditions without altering the underlying immersed boundary solver.

3. Active olfactory sampling induced by flapping flight

In this section, we investigate how flapping flight modifies near-field odor transport and enhances active sampling. We first compare the evolution of vortex and concentration field to identify the physical mechanism governing odor fluctuation. The proposed scalar jump-condition correction method is then validated using the computed concentration field around thin flapping wings. Finally, odor concentration measured by the virtual olfactory receptor array are compared with those obtained from a stationary wing reference case to quantify the enhancement in odor acquisition.

From fluid perspectives, odor molecules behave as a passive scalar whose transport is governed by the advection–diffusion equation

$$\frac{\partial C}{\partial t} + \vec{u} * \nabla C = D \nabla^2 C$$

where C is the odor concentration; u is the unsteady velocity field generated by wing flapping; D is the molecular diffusivity. Consequently, the periodic formation and shedding of vortical structures continuously redistribute odor within the near field.

Fig. 2 compares the instantaneous vorticity and scalar concentration fields on the mid-span section during a representative flapping cycle. The leading-edge vortex (LEV) and trailing-edge vortex (TEV) are generated alternately throughout the downstroke and upstroke. These vortices entrain the surrounding odor and transport it toward the antennal region, producing pronounced spatial and temporal variations in odor concentration.

The concentration field around thin flapping wings also provides a rigorous validation of the proposed scalar jump-condition correction method. Fig. 3(a) shows the instantaneous scalar concentration extracted on the central plane ($y=0$) passing through the wing mid-span. In theory, the impermeable wing surface requests sharp concentration gradients or even local discontinuities across the two sides of the interface. As illustrated in Fig. 3(b,c), the proposed jump-condition correction accurately preserves the concentration difference across the thin wing while maintaining overall numerical stability. The location of the concentration jump remains well aligned with the physical wing surface, demonstrating that the method effectively captures the prescribed boundary condition.

To quantify the influence of wing flapping on odor acquisition, the scalar concentration measured at the virtual olfactory probes are compared with those obtained from a stationary wing reference case. As shown in Fig. 4(a,b), the concentration signals under flapping flight exhibit pronounced periodic oscillations, while the stationary case produces relatively steady signals. More importantly, the time-averaged scalar concentration at all antennal probes is consistently higher under flapping conditions. These results indicate that wing flapping not only introduces temporal fluctuations but also enhances the amount of odor transported into the antennal region. Rather than passively sampling surrounding odor plume, mosquitoes actively reshape the surrounding flow field through wing motion, generating vortex entrainment that enhances near-field odor transport. The underlying transport mechanisms

responsible for this enhancement are further verified in the following sections using Lagrangian particle tracking and Lagrangian coherent structure analysis.

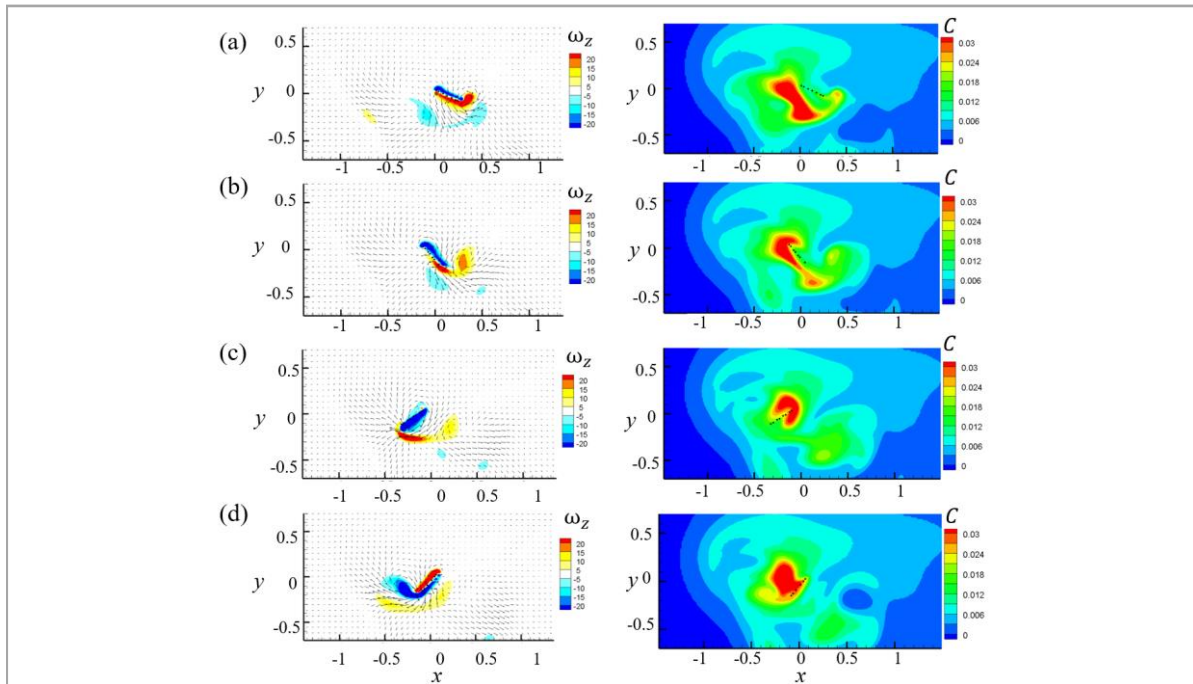


Figure 2. Instantaneous vorticity (left) and scalar concentration (right) fields on the spanwise mid-plane at four representative phases of the mosquito wingbeat cycle. (a) Early downstroke ($t/T=0.20$); (b) mid-downstroke ($t/T=0.40$); (c) early upstroke ($t/T=0.60$); and (d) mid-upstroke ($t/T=0.80$).

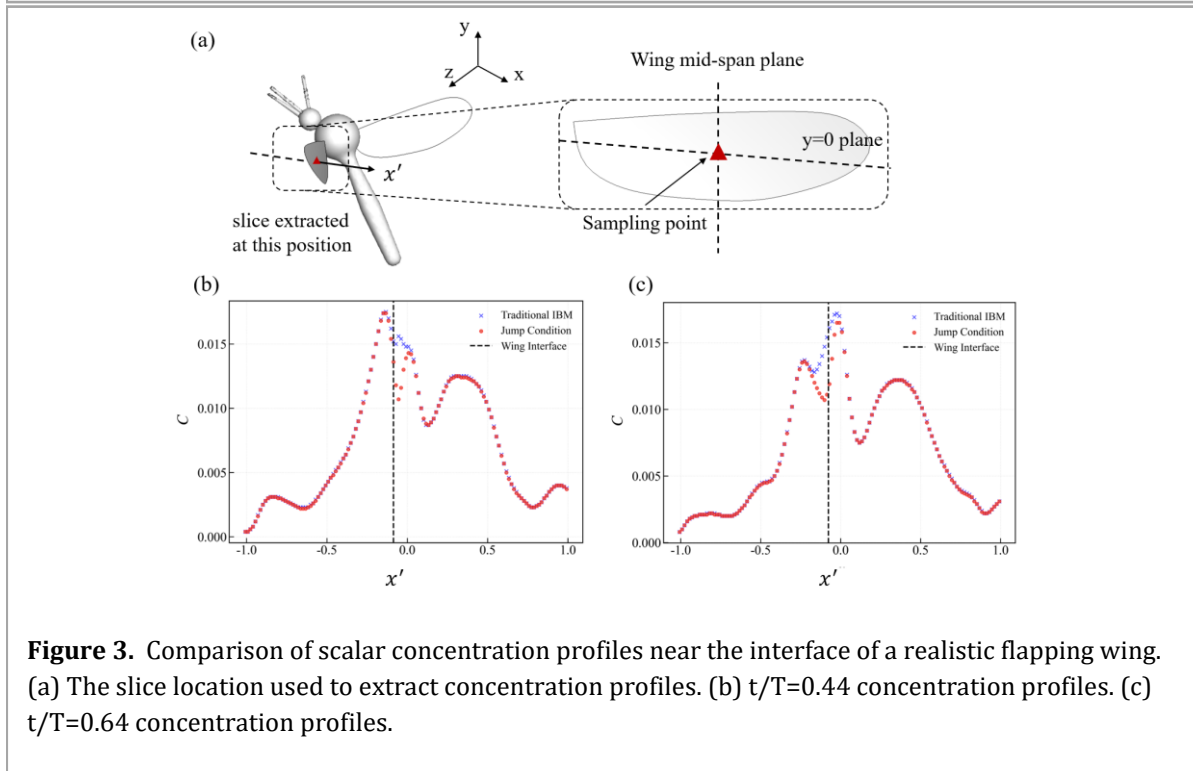


Figure 3. Comparison of scalar concentration profiles near the interface of a realistic flapping wing. (a) The slice location used to extract concentration profiles. (b) $t/T=0.44$ concentration profiles. (c) $t/T=0.64$ concentration profiles.

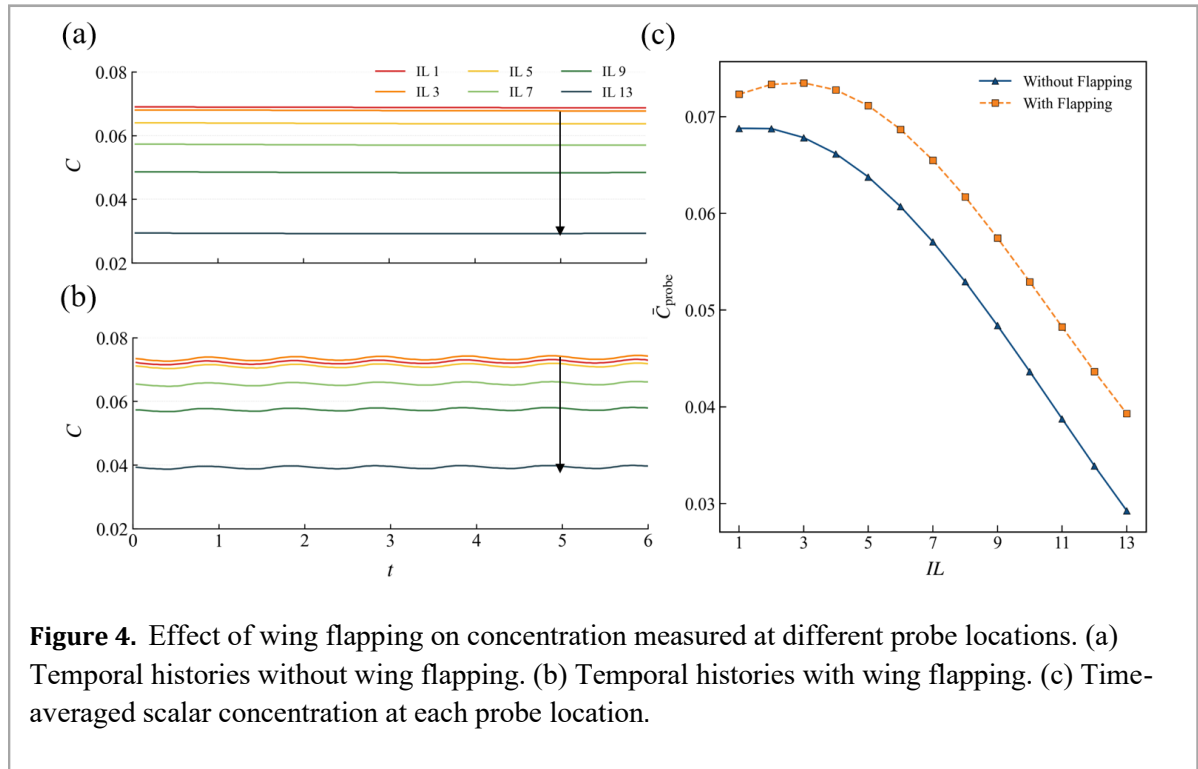


Figure 4. Effect of wing flapping on concentration measured at different probe locations. (a) Temporal histories without wing flapping. (b) Temporal histories with wing flapping. (c) Time-averaged scalar concentration at each probe location.

Following the Eulerian analysis of the scalar field, Lagrangian particle tracking was employed to provide a more intuitive description of flapping flight impact near field. Odor molecules were modelled as massless passive tracers whose trajectories satisfy

$$\frac{d\vec{x}}{dt} = \vec{u}(x, t)$$

where $\vec{u}(x, t)$ is the instantaneous velocity field. Particle trajectories are integrated using the classical fourth-order Runge–Kutta scheme.

Fig. 5 compares the trajectories of particles released from the upstream odor source with and without wing flapping. In the stationary case, the released particles exhibit the typical flow pattern of a bluff body, bypassing the mosquito body and remaining largely outside the antenna region. In contrast, wing flapping substantially modifies the near-field transport pathways. Particles are entrained toward the antennae and the mouthpart through the unsteady vortical flow. This enhanced transport pathway is consistent with the vortex-induced scalar accumulation observed in the previous section, providing a Lagrangian interpretation of the enhanced odor acquisition process.

To further clarify the transport mechanism, the particle trajectories are analyzed together with the instantaneous velocity fields on several representative planes, as shown in Fig. 6. On the $z=0$ (x - y) plane, the downstroke and upstroke generate alternating streamwise flows that periodically accelerate and decelerate the incoming flow, forming an entrainment-release cycle; on the $y=5\Delta x$ (x - z) plane, lateral transport remains weak, indicating that the flow is predominantly two-dimensional. Consequently, odor transport is primarily confined to the flapping plane; on the $x=-0.58$ (y - z) plane, the flapping wings alternately repel and entrain the surrounding fluid in the transverse direction. Together, the streamwise and transverse motions continuously renew the fluid near the antennae, while simultaneously generating localized stagnation regions.

The present study considers the flow generated by mosquito flapping flight, which is characterized by a relatively low Reynolds number. Under such conditions, viscous diffusion rapidly dominates the shed vortices, preventing the formation of long and coherent vortex streets typically observed in high Reynolds number flows [14]. Consequently, conventional Eulerian analysis based on instantaneous vorticity or wake structures alone is insufficient to reveal the transport pathways governing upstream–downstream fluid exchange. To overcome this limitation, Lagrangian coherent structures (LCS) are

employed to identify the attracting and repelling material structures that organize fluid transport [15–17]. In this study, LCS are identified using the finite-time Lyapunov exponent (FTLE), which characterizes the maximum stretching rate in a finite period of time [18].

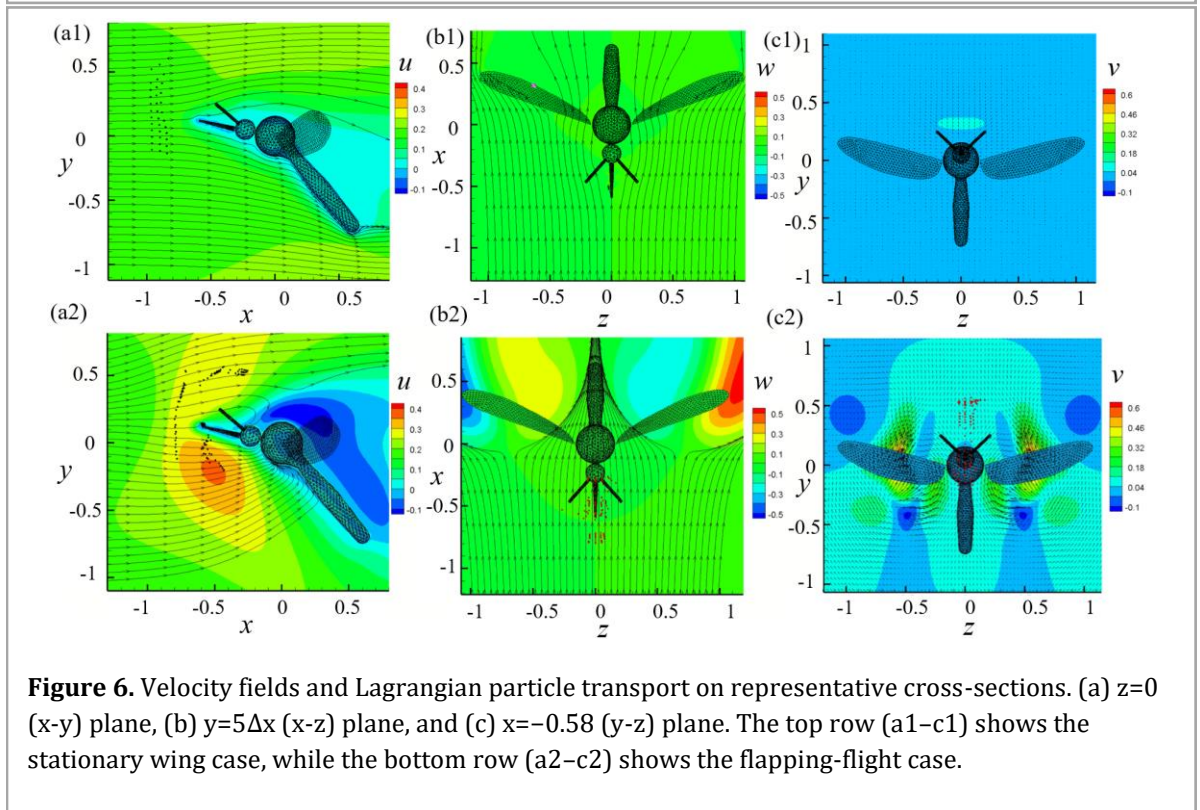
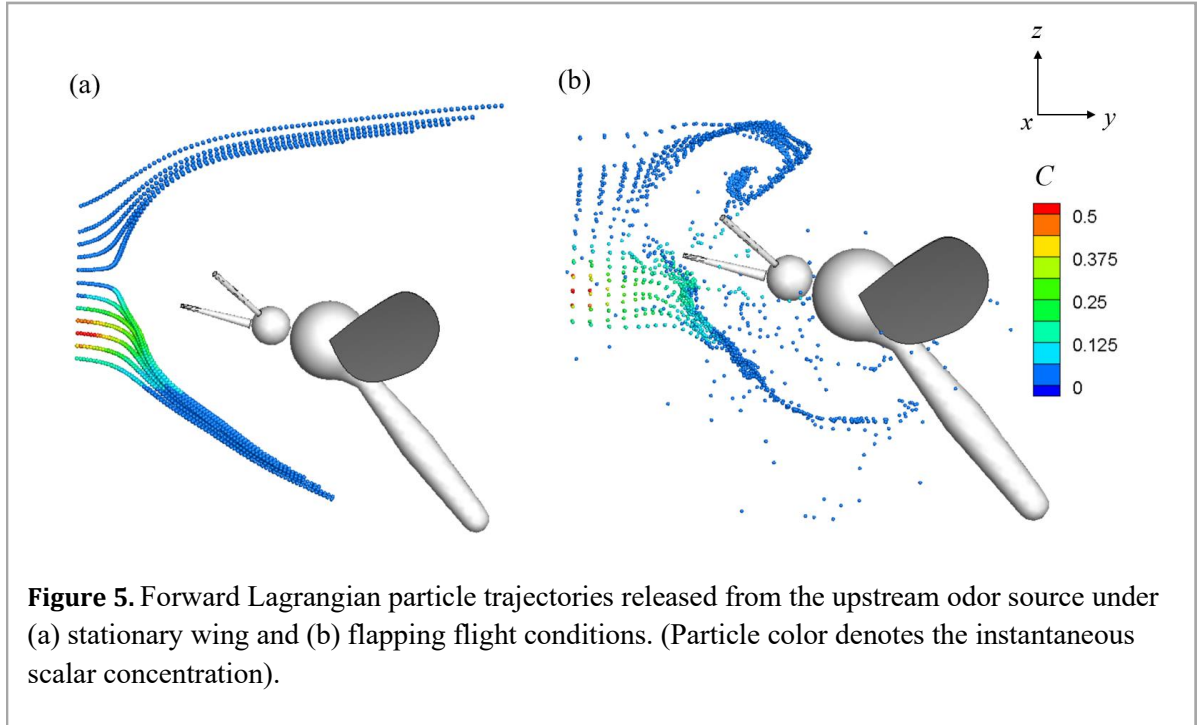


Fig. 7 presents the FTLE fields on representative cross-sections of the flapping flow. A layered FTLE pattern is observed around the wings, reflecting the intense and repeated shear generated during the flapping cycle. On the x - z section (Fig. 7b), the downwash induced by the wings forms two distinct

transport boundaries upstream of the body. Fluid on one side is entrained into the flapping-induced flow and undergoes strong shear, whereas fluid on the other side is diverted away from the body. On the x - y section (Fig. 7c), a pronounced FTLE ridge appears beneath the mouth, indicating the presence of a transport barrier that separates the incoming flow into two branches. One branch remains locally stagnant, while the other is entrained into the downstream transport pathway. This observation agrees well with the stagnation region identified from the Lagrangian particle trajectories (Fig. 5), providing further evidence that the near-field transport topology is fundamentally reorganized by wing flapping.

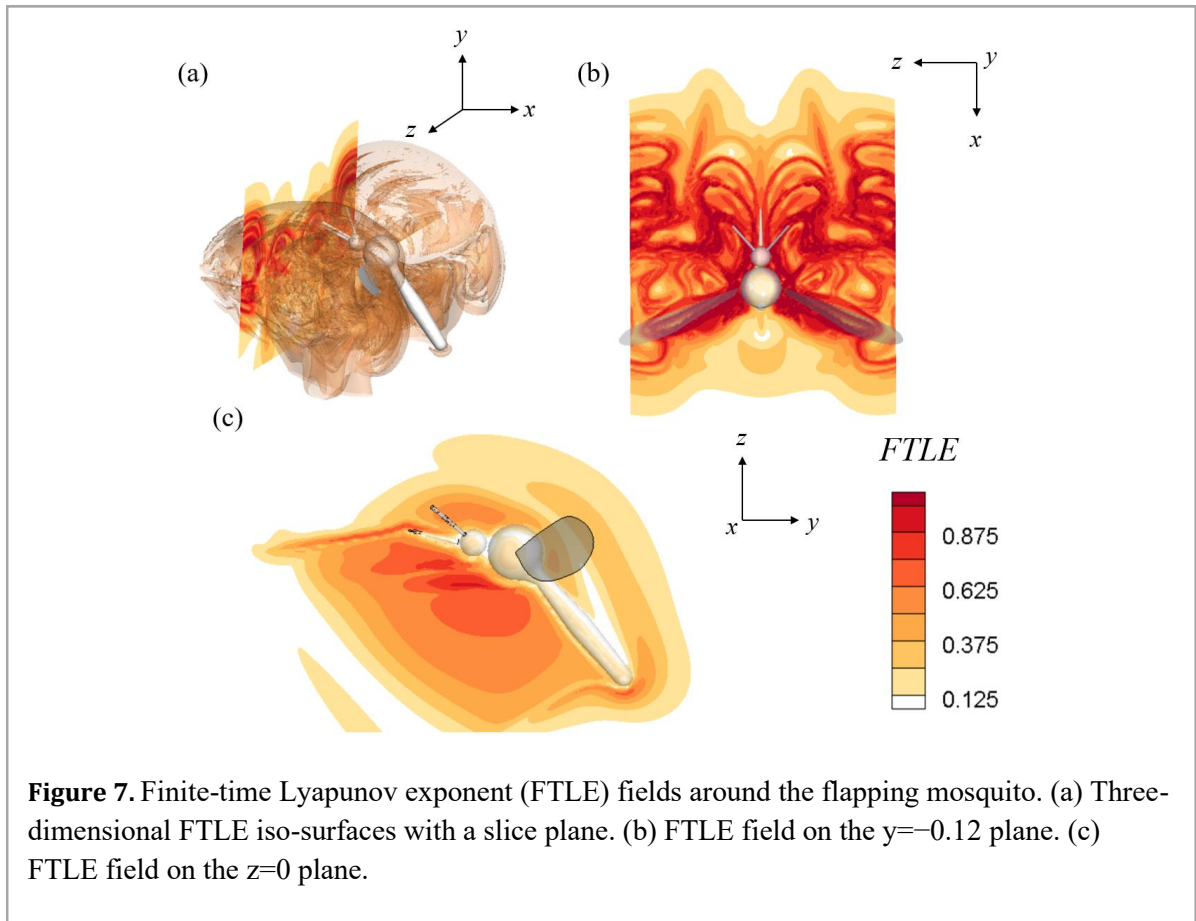


Figure 7. Finite-time Lyapunov exponent (FTLE) fields around the flapping mosquito. (a) Three-dimensional FTLE iso-surfaces with a slice plane. (b) FTLE field on the $y = -0.12$ plane. (c) FTLE field on the $z = 0$ plane.

4. Bio-inspired odor source localization

The previous sections focused on the forward problem, namely how wing flapping modifies near-field odor transport and the resulting concentration field. In practice, however, mosquitoes face the inverse problem: the location of an odor source is unknown and must be inferred solely from the concentration signals detected by the antennal receptors. This naturally raises the following question: to what extent can the source location be reconstructed from the measured concentration field under flapping-induced impact?

In studies of atmospheric dispersion and biological olfactory navigation, the Gaussian plume model has been widely adopted as a simplified description of scalar transport and serves as the basis for many odor-source localization strategies [19-22]. The classical Gaussian plume solution is given by

$$C(x, y, z) = \frac{Q}{2\pi u_r \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left[\exp\left(-\frac{(z-h)^2}{2\sigma_z^2}\right) + \alpha \exp\left(-\frac{(z+h)^2}{2\sigma_z^2}\right) \right]$$

To evaluate whether this widely used model remains applicable in the presence of flapping-induced near-field transport, a simplified three-dimensional Gaussian plume model was first constructed,

$$C = A * \frac{L_{smooth}}{L_{smooth} + (x - x_c)} \exp \left[-\frac{(y - y_c)^2 + (z - z_c)^2}{2 * D_{coeff} * \frac{x - x_c}{U_{bp}}} \right]$$

where C is the scalar concentration at the sensing location; A is the amplitude, D_{coeff} is the effective diffusion coefficient; U_{bp} is the characteristic convection velocity; L_{smooth} is a near-field smooth parameter; and (x_c, y_c, z_c) represents the odor source location. The model parameters were calibrated using the reference configuration without source displacement and subsequently evaluated for odor sources located at different spatial positions.

Algorithm 1. Gaussian plume surrogate model pipeline

Input: Sensor locations $\{x_k\}_{k=1}^{26}$, CFD-derived sensor concentrations $\{C_k^{CFD}\}_{k=1}^{26}$.

Output: Calibrated Gaussian-model parameters D_{coeff}^* and A^* , and the inferred source displacement $\Delta^* = (\Delta x^*, \Delta y^*, \Delta z^*)^T$.

Stage 1. Calibration using the no-shift case

Given the concentrations measured at the 26 sensor locations from the reference CFD simulation, and determine the optimal model parameters by minimizing

$$(D_{coeff}^*, A^*) = \arg \min_{D_{coeff}, A} \sum_{k=1}^{26} (C_k^{CFD} - C_k^{Gauss}(D_{coeff}, A))^2.$$

Stage 2. Source location inference

For a shifted source, compute the Gaussian surrogate concentrations using the calibrated parameters (D_{coeff}^*, A^*)

$$C_k^{Gauss} = \mathcal{G}(x_k; D_{coeff}^*, A^*, \Delta x, \Delta y, \Delta z),$$

and estimate the source displacement by minimizing

$$(\Delta x^*, \Delta y^*, \Delta z^*) = \arg \min_{\Delta x, \Delta y, \Delta z} \sum_{k=1}^{26} (C_k^{CFD} - C_k^{Gauss}(D_{coeff}^*, A^*, \Delta x, \Delta y, \Delta z))^2.$$

Fig. 8 compares the concentration profiles predicted by the Gaussian plume model with those obtained from the CFD simulations. Overall, the Gaussian model successfully captures the large spatial trends of the concentration field. For lateral source displacements, it correctly predicts the concentration difference between the left and right antennae. For vertical displacements, it also predicts the overall increase or decrease concentration associated with the shift of the plume centerline. However, significant discrepancies appear in the near field. As a time-averaged analytical model, the Gaussian plume neglects the flow modulation generated by wing flapping. Consequently, it fails to reproduce the local nonlinear variations in concentration induced by vortex entrainment and shear, leading to noticeable deviations from the CFD results, particularly along the antennal sensing array.

To further quantify the influence of these nonlinear transport effects on source localization, we plot errors predicted by Gaussian model for different source positions. As shown in Fig.9, localization error increases substantially when near-field impact becomes significant, demonstrating the nonlinear scalar transport induced by wing flapping cannot be adequately represented by a conventional Gaussian plume model with constant diffusion parameters.

Motivated by these observations, a neural network surrogate model is introduced in the following section to learn the non-linear mapping between concentration signals and odor source locations. Rather than learning this mapping solely from CFD data, a transfer learning framework is adopted to improve both learning efficiency and model generalization. The underlying idea is to exploit the analytical Gaussian model to capture the global spatial mapping, while a limited amount of high-fidelity CFD data is subsequently used to capture the local nonlinear effects introduced by flapping-induced flow modulation. The training procedure therefore consists of two stages: first, the neural network is pretrained using synthetic data generated from the Gaussian plume model. The pretrained network is then fine-tuned using CFD-derived concentration data.

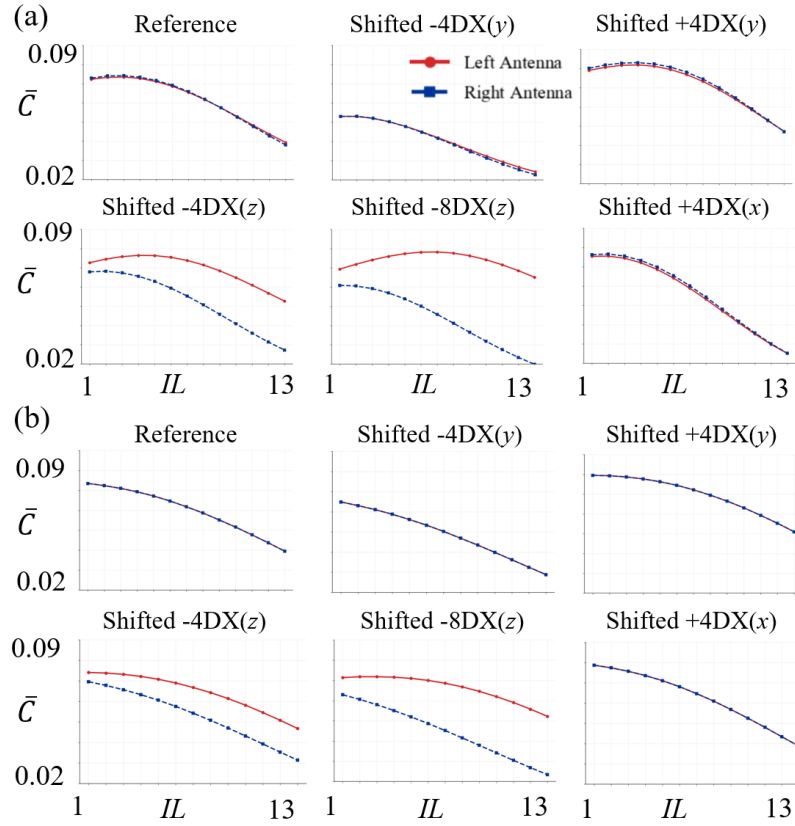
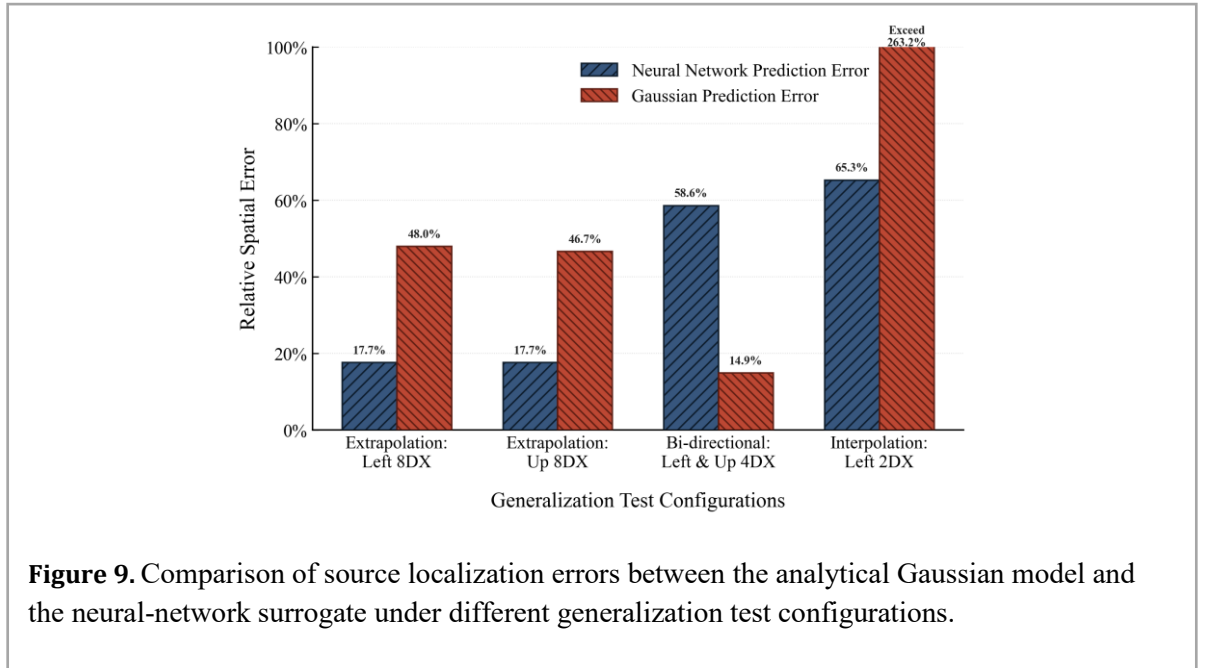


Figure 8. Time-averaged concentration distribution along the left and right antennae under multidirectional odor-source displacements for (a) the realistic flapping-wing model and (b) Ideal Gaussian plume.

To evaluate the generalization capability of the proposed model, four independent test cases excluded from the training dataset are considered, including interpolation, extrapolation and simultaneous two-dimensional source displacements. The localization errors are summarized in Fig. 9 (blue bars). The proposed model demonstrates good generalization for large source displacements. In contrast, the prediction accuracy decreases for interpolation and multi-directional displacement cases, where stronger local nonlinearities dominate in the near field. Consequently, relatively small changes in source location can result in pronounced variations in the measured concentration signals, increasing the difficulty of the inverse mapping. Further improvements may be achieved by expanding the training dataset, adopting deeper network architectures, or incorporating governing physical constraints through physics-informed neural networks (PINNs). These strategies are expected to further improve both localization accuracy and model generalization under complex flapping-induced flow conditions.



Algorithm 2. Physics-guided neural network surrogate model pipeline

Input: Gaussian plume model dataset $\mathcal{D}_G$; high-fidelity CFD dataset $\mathcal{D}_C$.

Output: Trained neural network f_θ for predicting the source displacement $(\Delta x, \Delta y, \Delta z)$.

Stage 1. Data preprocessing

Extract concentration features from the 26 probe channels to form the network input

$$\mathbf{x} = [f_{LR}, f_{Mean}, f_{TB}]^T,$$

which serve as the network input.

Stage 2. Neural network construction

Construct a fully connected neural network

$$\hat{\mathbf{y}} = f_\theta(\mathbf{x}) = (\Delta x, \Delta y, \Delta z)^T.$$

Stage 3. Physics-guided transfer learning

Gaussian prior pre-training. Initialize the network using the Gaussian surrogate dataset $\mathcal{D}_G$ by minimizing

$$L_G = \frac{1}{N_G} \sum_{i=1}^{N_G} \|f_\theta(\mathbf{x}_i) - \mathbf{y}_i\|_2^2,$$

yielding the pretrained parameters θ_G^* .

CFD-based fine tuning. Initialize $\theta \leftarrow \theta_G^*$ and optimize

$$L_C = \frac{1}{N_C} \sum_{i=1}^{N_C} \|f_\theta(\mathbf{x}_i) - \mathbf{y}_i\|_2^2,$$

using the CFD dataset $\mathcal{D}_C$ to obtain the final parameters θ^* .

Stage 4. Source location inference

For an unseen concentration measurement, the model predicts the source displacement

$$(\Delta x, \Delta y, \Delta z) = f_{\theta^*}(\mathbf{x}^{obs}).$$

5. Conclusions

This study investigated active olfactory sampling in flapping mosquitoes using a three-dimensional numerical framework coupling unsteady aerodynamics with passive scalar transport. To

accurately resolve scalar transport around thin flapping wings, a scalar jump-condition correction method was developed within the immersed boundary framework. Based on this computational framework, both the forward problem of odor transport and the inverse problem of odor-source localization were systematically investigated.

First, wing flapping fundamentally reorganizes near-field odor transport. Periodic formation and shedding of leading-edge and trailing-edge vortices continuously entrain surrounding fluid, producing both stronger temporal concentration fluctuations and higher time-averaged concentrations at the antennal sensing locations. Lagrangian particle tracking further reveals that wing flapping redirects odor transport toward the antennae, while FTLE analysis identifies coherent transport barriers that organize upstream–downstream fluid exchange. Second, flapping-induced nonlinear transport substantially influences inversed odor source localization. And we adopt a new physics-guided neural network surrogate model to successfully capture the nonlinear relationship between concentration measurements and odor-source locations.

Although the present study provides new insights into the fluid mechanical basis of active olfactory sampling in mosquitoes, several limitations remain. First, the computational domain is constrained by the available computational resources. At higher flapping frequencies, stronger unsteady wake structures may interact with the domain boundaries and influence the near-field flow. Second, the proposed inverse-sensing framework was developed and validated using numerical data. Future work will incorporate experimental measurements and learning-based navigation strategies to establish a closed-loop framework linking odor sensing, environmental perception, and autonomous navigation.

Acknowledgements

Appendix

Weak form derivation of the scalar jump condition forcing term

To construct a forcing formulation capable of enforcing scalar jump conditions within the immersed boundary framework, we consider the advection–diffusion equation augmented by an interfacial source term,

$$\frac{\partial C}{\partial t} + u \cdot \nabla C = D \nabla^2 C + f \quad (\text{A-1})$$

$$f = \iint F(r, s, t) \delta(x - X(r, s, t)) dr ds \quad (\text{A-2})$$

Because the scalar concentration and pressure derivative may not exhibit well definition near the interface. Instead, a weak formulation is employed to derive the corresponding jump conditions. Consider a thin neighborhood $\Omega_\varepsilon(t)$ enclosing the interface with thickness ε , and introduce an arbitrary smooth test function φ . Integrating Eq.(A1) over $\Omega_\varepsilon(t)$ gives

$$\begin{aligned} \int_{\Omega_\varepsilon(t)} \left(\frac{\partial C}{\partial t} + u \cdot \nabla C \right) \varphi dx &= \int_{\Omega_\varepsilon(t)} D \Delta C \varphi dx \\ &+ \int_{\Omega_\varepsilon(t)} [\iint F(r, s, t) \delta(x - X(r, s, t)) dr ds] \varphi dx \end{aligned} \quad (\text{A-3})$$

The left-hand side can be written as

$$LHS = \frac{d}{dt} \int_{\Omega_\varepsilon(t)} C \varphi dx \quad (\text{A-4})$$

Since the scalar concentration remains bounded on both sides of the interface, when

$$\varepsilon \rightarrow 0, \quad LHS \rightarrow 0$$

The two terms on the right-hand side are treated separately.

Diffusion term

Applying Green's formula,

$$\begin{aligned} \iint_{\Omega} (\nabla \cdot G) \varphi dx dy &= \int_{\partial \Omega} (G \cdot n) \varphi ds - \iint_{\Omega} G \cdot \nabla \varphi dx dy \\ RHS1 &= \int_{\Omega \varepsilon(t)} D \Delta C \varphi dx \\ &= D \int_{\partial \Omega \varepsilon(t)} (\nabla C \varphi) \cdot n dl - D \int_{\Omega \varepsilon(t)} \nabla C \cdot \nabla \varphi dx \\ &= D \int_{\partial \Omega \varepsilon(t)} (\nabla C \varphi) \cdot n dl - D \left[\int_{\Omega \varepsilon(t)} \nabla \cdot (C \nabla \varphi) dx - \int_{\Omega \varepsilon(t)} C \cdot \Delta \varphi dx \right] \end{aligned} \quad (A-5)$$

As $\varepsilon \rightarrow 0$, C remains bounded while the volume integral vanishes. Only the interface contribution remains,

$$RHS1 = D \iint [\nabla C] \varphi \cdot n \left| \frac{\partial X}{\partial r} \times \frac{\partial X}{\partial s} \right| dr ds - D \iint [C] \nabla \varphi \cdot n \left| \frac{\partial X}{\partial r} \times \frac{\partial X}{\partial s} \right| dr ds \quad (A-6)$$

Interfacial forcing term

$$\begin{aligned} RHS2 &= \int_{\Omega \varepsilon(t)} \left[\iint F(r, s, t) \delta(x - X(r, s, t)) dr ds \right] \varphi dx \\ &= \iint F(r, s, t) \left[\int_{\Omega \varepsilon(t)} \delta(x - X(r, s, t)) \varphi dx \right] dr ds \\ &= \iint F(r, s, t) \left[\int_{\Omega \varepsilon(t)} \varphi(X(r, s, t)) dx \right] dr ds \end{aligned} \quad (A-7)$$

Combining Eqs. (A3), (A6), and (A7) leads to the interfacial jump relation,

$$[\nabla C] \varphi \cdot n \left| \frac{\partial X}{\partial r} \times \frac{\partial X}{\partial s} \right| - [C] \nabla \varphi \cdot n \left| \frac{\partial X}{\partial r} \times \frac{\partial X}{\partial s} \right| = F(r, s, t) \varphi \quad (A-8)$$

$$f = \iint F1 \delta(x - X) dr ds + \iint F2 \nabla \delta(x - X) \cdot n dr ds \quad (A-9)$$

where

$$F1 = [\nabla C_n] \left| \frac{\partial X}{\partial r} \times \frac{\partial X}{\partial s} \right| \quad (A-10)$$

$$F2 = D[C] \left| \frac{\partial X}{\partial r} \times \frac{\partial X}{\partial s} \right| \quad (A-11)$$

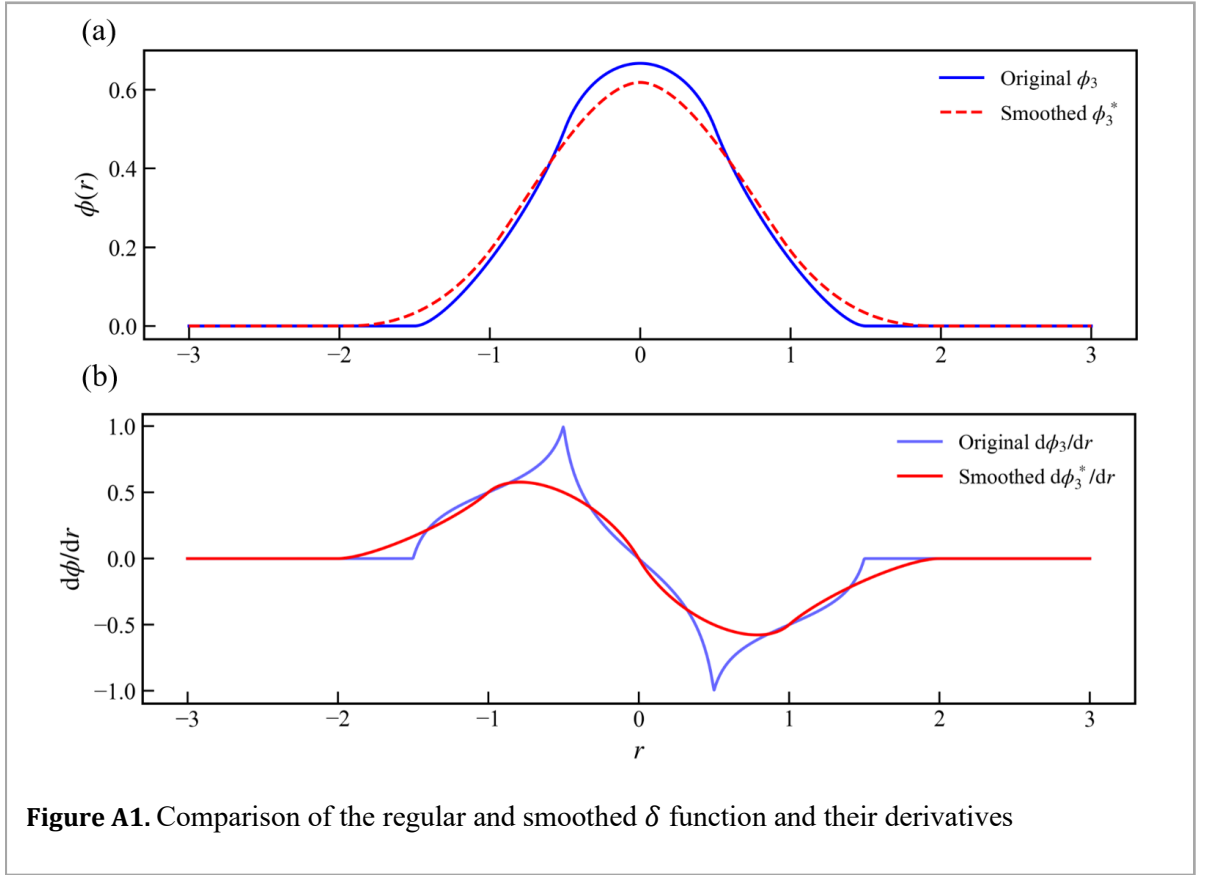


Figure A1. Comparison of the regular and smoothed δ function and their derivatives

Since the proposed formulation involves both the discrete δ function and its derivative, direct discretization may introduce numerical oscillations near the immersed interface. To improve numerical stability, the smoothing function proposed by Yang *et al.* [22] was adopted. The discrete kernel function and its derivative were regularized according to

$$\varphi^*(r) = \int_{r-0.5}^{r+0.5} \varphi(r') dr'$$

with the corresponding derivative written as

$$\nabla \varphi^*(r) = \varphi^*(r + 0.5) - \varphi^*(r - 0.5)$$

The smoothed kernel function and its gradient were illustrated in Fig. A1. To further validate the capability of the proposed jump-condition correction method for imposing discontinuous boundary conditions, the irregular interface diffusion benchmark by Gibou *et al.* [27] was employed. The governing equation is the two-dimensional diffusion equation,

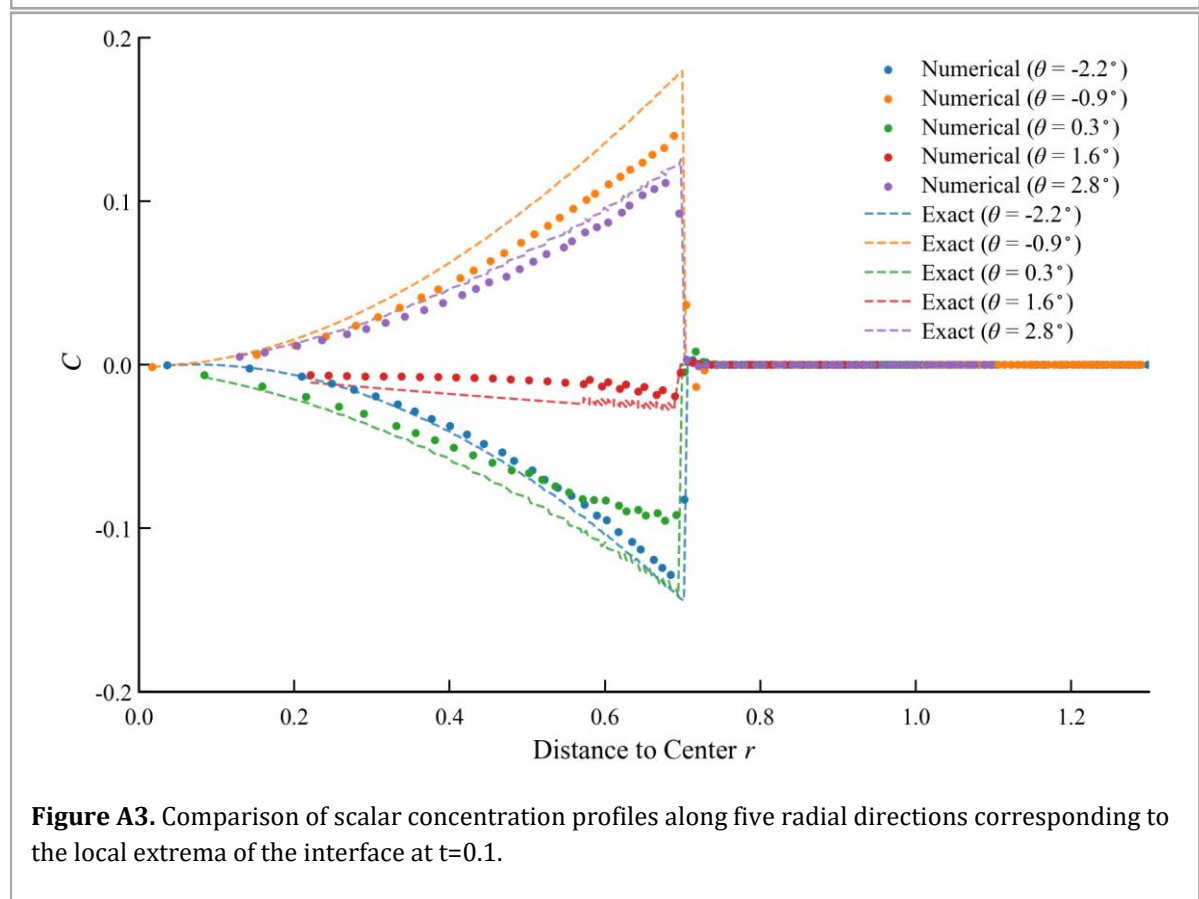
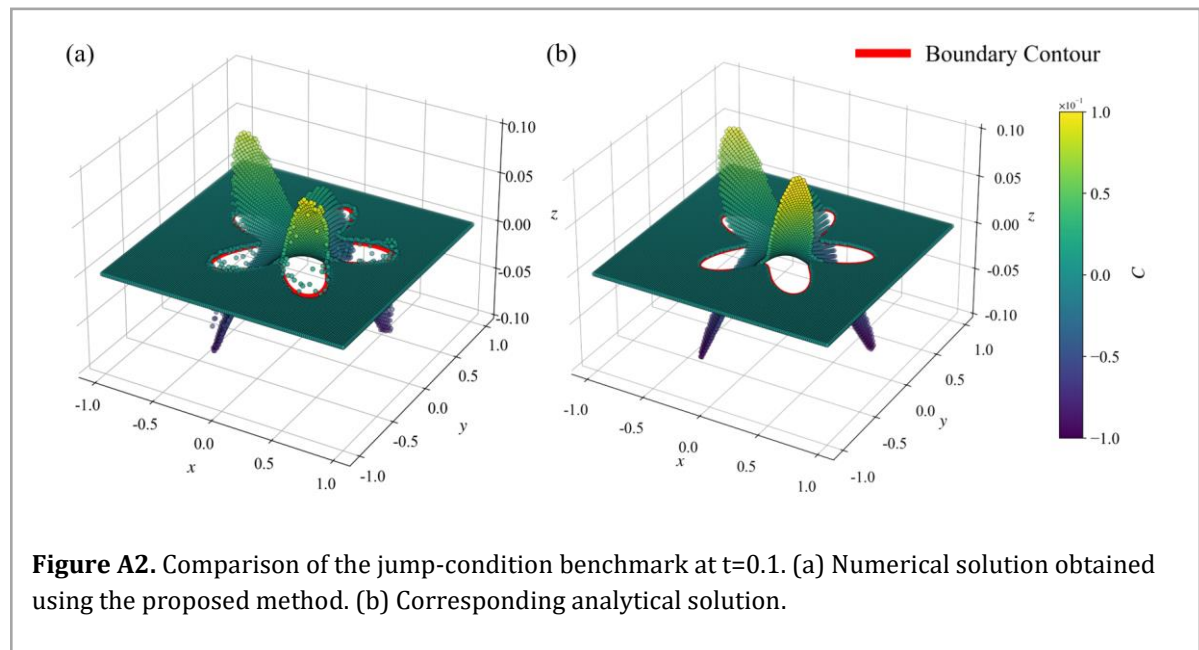
$$Tt = \Delta T \quad \text{in } \Omega = [-1,1] \times [-1,1]$$

where the immersed interface is described parametrically by

$$\begin{aligned} x(\theta) &= 0.02\sqrt{5} + (0.5 + 0.2 \sin(5\theta)) \cos(\theta) \\ y(\theta) &= 0.02\sqrt{5} + (0.5 + 0.2 \sin(5\theta)) \sin(\theta) \end{aligned}$$

The analytical solution provides both the prescribed scalar jump $[T]$, and the jump in the normal gradient $[T_n]$ across the interface. These jump conditions were used to evaluate the monopole forcing term $F1$ and the dipole forcing term $F2$ according to Eqs. (A-10) and (A-11).

Fig. A2 compares the numerical solution obtained using the present method with the analytical solution. Well agreement is observed for the scalar discontinuity across the immersed interface. In particular, the proposed jump-condition correction accurately preserves the sharp interfacial jump without introducing noticeable numerical oscillations.



To quantitatively validate the proposed method, scalar profiles were extracted along five radial directions passing through the local maxima of the immersed interface and compared with the analytical solution, as shown in Fig. A3. The results demonstrate that the inclusion of the dipole forcing term enables the present method to accurately recover the interfacial discontinuity, producing numerical profiles that agree closely with the analytical solution near the immersed boundary.

It is also observed that the theoretical discontinuity is not reconstructed as ideal but rather appears as a thin transition layer with a finite thickness. This behavior is an inherent characteristic of immersed boundary methods, in which the forcing is distributed from the Lagrangian interface to neighbour Eulerian grid points through a discrete δ function. Consequently, the interface is represented over several grid cells rather than as a mathematically sharp discontinuity. Nevertheless, the proposed jump-condition correction still improves the representation of discontinuous boundary conditions while preserving the robustness and numerical stability of the immersed boundary framework.

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